

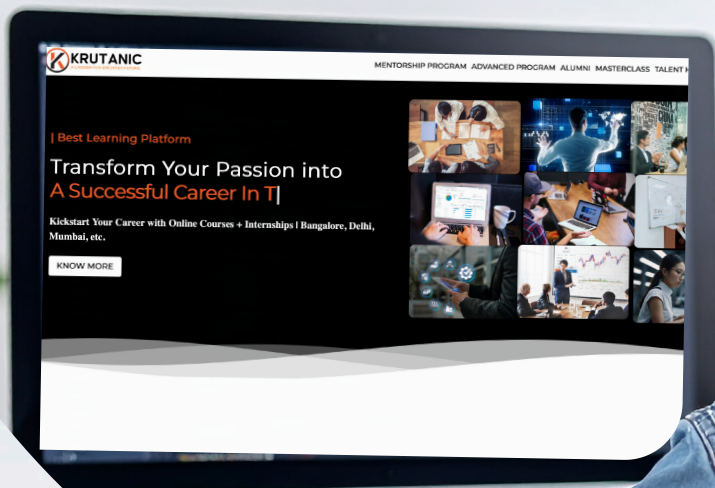


KRUTANIC

A LADDER FOR BRIGHTER FUTURE

*Transform Your Passion into
A Successful Career In Tech*

VLSI Design



We are now an **accredited**
partner under



www.krutanic.com



support@krutanic.com



VLSI Design

ABOUT US

Krutanic is at the forefront of transforming education through cutting-edge technology. Our comprehensive platform empowers learners with personalized learning experiences, collaborative tools, and real-time analytics. With adaptive assessments and interactive content creation, we enhance student engagement and achievement. Join us in revolutionizing education for the digital age, driving positive outcomes and preparing learners for success in tomorrow's world.

TRAINING AND INTERNSHIP PROGRAM

Why VLSI Design?

- Backbone of modern chips and electronics
- High demand in semiconductor and tech industries
- Multiple career paths: design, verification, PD, low-power
- Focus on speed, power, and area optimization
- Strong industry and real-chip relevance
- High-growth, well-paying global careers

TRAINING AND INTERNSHIP PROGRAM

1st MONTH



Live Expert-Led Sessions

- Interactive live sessions conducted by industry veterans with 10+ years of hands-on experience in semiconductor product and service companies.
- Sessions are driven by real project scenarios, design challenges, and best practices followed in production environments.



Session Recordings & LMS Access

- All live sessions are recorded and archived with 1-year unlimited access through our dedicated LMS portal.
- Learners can revisit complex topics, revise before interviews, and continue learning at their own pace.



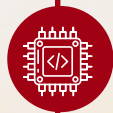
Industry Aligned Curriculum

- Curriculum is designed and reviewed by senior working professionals to ensure alignment with current industry workflows, tools, and expectations.
- Focus is on what engineers actually do on the job, not just theoretical concepts.



Hands-On, Tool-Driven Learning

- Extensive hands-on exposure to industry-standard EDA tools used across RTL design, synthesis, timing analysis, and physical design.
- Learners gain practical confidence in navigating real tool environments and design flows.



End-to-End ASIC Design Flow Exposure

- Structured understanding of the complete VLSI design lifecycle from RTL development and verification to synthesis and backend implementation concepts enabling learners to see the “big picture” of chip development.



Real-World Design Assignments

- Assignments and exercises modeled on actual industry use cases, enabling learners to apply concepts to realistic design problems rather than academic examples.



Industry Coding & Design Best Practices

TRAINING AND INTERNSHIP PROGRAM

2nd MONTH



Real-Time Industry Projects & Expert Mentorship

- Industry Grade Real Time Projects
Learners gain hands-on experience through two carefully designed real-time industrial projects:
- One Minor Project to strengthen core fundamentals and tool flow understanding
- One Major Project that simulates a full-scale industry design cycle



Mentor-Led Project Execution

- Every learner is guided by senior industry mentors acting as project leads, offering structured support from project kickoff to final delivery.
- This mentor-driven approach closely mirrors professional engineering team environments.



End-to-End Project Exposure

Projects follow an industry-standard execution model, including:

- Requirement analysis & design planning
- Architecture discussion and implementation
- Debugging, optimization, and validation
- Final review and project sign-off
- Learners develop a clear understanding of real-world design ownership.



Hands-On Tool-Based Implementation

- All project work is executed using industry-standard EDA tools, enabling learners to gain confidence in professional tool environments and workflows expected in real projects.



Personal Development & Skill Enhancement

- Additional mentor-recommended mini-projects or enhancements may be assigned to support individual growth, specialization readiness, and technical depth.



Design Reviews & Technical Evaluation

Projects include formal design reviews and milestone evaluations, helping learners build:

- Technical communication skills
- Design justification ability
- Confidence in handling review-based feedback

Modules Overview



**CMOS Basics
(Recap)**

**Introduction to VLSI
Design**

**Combinational &
Sequential Logic
Optimization**

**Finite State Machines
(FSM) Mealy & Moore**

**Timing Concepts
(Setup, Hold, Skew, Jitter)**

**Verilog RTL Coding Best
Practices**

**SystemVerilog
Introduction**

**Synthesizable vs
Test bench Code**

RTL Design Flow

**Synthesis Process &
Constraints**

Physical Design (PD)

**Capstone Project &
Interview Preparation**

(only for placement assistance
program mock interview preparation)



Curriculum included

1

Introduction to VLSI Design

- Overview of VLSI design flow
- Frontend and backend concepts
- Chip design basics

2

CMOS Basics (Recap)

- CMOS inverter operation
- Switching behavior and noise margins
- Power dissipation basics

3

Combinational & Sequential Logic Optimization

- Reduce logic depth
- Improve speed and area
- Minimize power

4

Finite State Machines (FSM) – Mealy & Moore

- Control logic design
- State transitions and outputs
- Mealy vs Moore timing



Curriculum included

5

Timing Concepts (Setup, Hold, Skew, Jitter)

- Clock timing requirements
- Prevent timing violations
- Reliable data transfer

6

Verilog RTL Coding Best Practices

- Clock timing requirements
- Prevent timing violations
- Reliable data transfer

7

SystemVerilog Introduction

- logic, always_ff, always_comb
- Interfaces and testbench features
- Better design structure

8

Synthesizable vs Testbench Code

- Hardware-realizable code
- Simulation-only constructs
- Verification support



Curriculum included

9

RTL Design Flow

- RTL to synthesis flow
- Design implementation steps
- Frontend overview

10

Synthesis Process & Constraints

- RTL to gate-level netlist
- Timing, area, power constraints
- Design optimization

11

Physical Design (PD)

- Floorplanning and routing
- CTS and power planning
- Timing closure, DRC, LVS

12

Capstone Project & Interview Preparation

- Hands-on project work
- Practical VLSI exposure
- Interview readiness

Tools Included



systemverilog



QuestaSim / VCS



Design Compiler



PrimeTime



SpyGlass

Discover why this **VLSI Design** course is essential for your learning journey.

STEP 1



Google
Ratings 4.8/5

STEP 2



Mentees
Trained 15k+

STEP 3



Hiring
Partners 200+

STEP 6



Global Size USD
200 BILLION

STEP 5



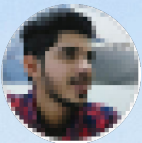
Average Salary
10+ LPA

STEP 4



Job Openings
25000+

Students Reviews



Ramshad K

★★★★★

I have gained a really hands on Experience with on graphic design which was delivered by Akash sir was very supportive and endless worthy experience so far.



Rajendra Prasad

★★★★★

I'm Rajendran Prasad, to be honest I'm expressing my words, I have researched and found the correct certificate on LinkedIn and scan that and found it 100% genuine and verified successfully.

It's my individual recommendation for everyone that Krutanic company is perfect with quality training and real world projects.

I have seen some bad reviews and many good reviews so got confused but now I realised that Krutanic solutions is the best company ever for training and internship provider, I have learned a lot of new industrial things.



Sahad K

★★★★★

I got placement at Wipro through krutanic support team. I am really grateful that I got to learn from AI Mentor Ashish sir and support from my counsellor was really helpful. All thanks to Bhoomika mam my operations support member from krutanic.



Sunil Kumar

★★★★★

I'm Sunil Kumar, I have enrolled for Data Science program at Krutanic solutions and my experience is outstanding.

It's my individual recommendation for everyone that Krutanic company is perfect with quality training and real world projects.



Shubham Rai

★★★★★

Krutanic Group is truly doing a remarkable job by creating learning spaces that are not only informative but also inspiring. I've had the opportunity to attend two of their courses-Photography and Graphic Designing-and both were simply

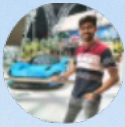


Aman Nema

★★★★★

Krutanics Full Stack Web Development program was a fantastic learning experience. The course covered essential technologies like HTML, CSS, JavaScript, React, and backend development, providing strong foundation for building real-world web applications. The hands-on projects and expert guidance from mentors made learning interactive and practical. This program is perfect for beginners and aspiring developers looking to enhance their skills in web development. Highly recommended.

Students Reviews



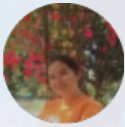
Tejas Kolekar

I am thrilled to share my enriching experience from the Artificial Intelligence and Machine Learning Internship I recently completed with Krutanic, in collaboration with Wipro, a leading multinational corporation. This 3-month program was an incredible opportunity to deepen my knowledge, enhance my skills, and grow both personally and professionally in the dynamic field of AI and ML.



Pranali Shinde

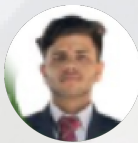
Excited to share that I've just completed a Data Science course from Krutanic. It was an amazing journey diving into data visualization, exploratory data analysis, SQL, and machine learning. I've gained some valuable skills that I can't wait to apply in real-world projects. Big thanks to the instructors and everyone who made this experience so enriching! Looking forward to what's next!



Prabhleen Kaur

Just after few months of completion of my AI course I got my hands on attending Master class on Building AI agent without coding held in the month of March my experience was very good I get to know about new things from my mentor Mr.Aashish Mishra And at the end i recieved this ~

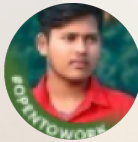
All thanks to Krutanic for giving me this opportunity.....



Rishabh Nagar

Certificate of Training Completion from Krutanic Solutions in the field of Android App Development Krutanic Paidi Venkata

[#krutanic](#) [#android](#) [#app](#)
[#development](#) [#androidstudio](#)



Himendra Shukla

Completing my internship at **KrutaniC** has been a transformative journey, equipping me with hands-on experience in **Full Stack Web Development** and preparing me for a bright future in tech. Grateful for the guidance, real-world projects, and skills gained— here's to innovation and success ahead! ★

[#InternshipCompletion](#) [#FullStackDevelopment](#) [#KrutaniC](#)
[#TechJourney](#)
[Krutanic](#)



Touheed Ahmed Nadaf

Clouds conquered.
Excited to share my Cloud Computing internship certification from Krutanic!

Grateful for the hands-on learning and ready to turn this experience into an impact in the ever-evolving tech landscape.

[#CloudComputing](#) [#Krutanic](#) [#InternshipWin](#)
[#TechReady](#) [#NextGenTech](#)
[#FutureInCloud](#) [#LevelUp](#) [#DigitalSkills](#)

Career Opportunities in VLSI Design

RTL Design Engineer Designs and codes synthesizable RTL using Verilog/System Verilog for digital blocks and controllers.

ASIC Design Engineer Works on CMOS-based digital design, logic optimization, FSMs, and timing-aware circuit design.

Synthesis Engineer Converts RTL into gate-level netlists using synthesis tools while meeting timing, area, and power constraints.

Physical Design (PD) Engineer Handles floor planning, placement, routing, CTS, and timing closure to produce layout-ready designs.

Verification Engineer Ensures RTL correctness using functional verification, assertions, and UVM-based environments.

Low-Power VLSI Engineer Implements power gating, multi-Vt cells, and DVFS techniques to reduce power consumption in chips.

Certifications

01

**Training
Completion
Certificate of
Validates the skills
which acquired**

02

**Internship
Completion
certificate
certified by
Adobe**

04

**Certificate of
excellence
(Based on you
Performance)**

03

**Letter of
recommendation(L
OR) for your
JOB/PLACEMENT**

STUDENT AVERAGE RATINGS

★★★★★ 4.85/5

CERTIFICATE OF TRAINING COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Danish Raja Akhtar

Your dedication and achievement are truly inspiring, reflecting your readiness to excel in the dynamic field. This Certificate of Training Completion is awarded for successfully completing the

Full Stack Web Development on May 2025



Shrikant
SHRIKANT RATHOD
EXECUTIVE DIRECTOR



CERTIFICATE OF INTERNSHIP

May 5th, 2025 to July 5th, 2025

This Certificate is Proudly Presented To

Danish Raja Akhtar

This is to certify that the candidate has successfully completed the Internship. During the internship period, they demonstrated strong technical skills, professionalism, and a keen interest in the field. Their contributions were valuable to the team in the domain of **Full Stack Web Development**



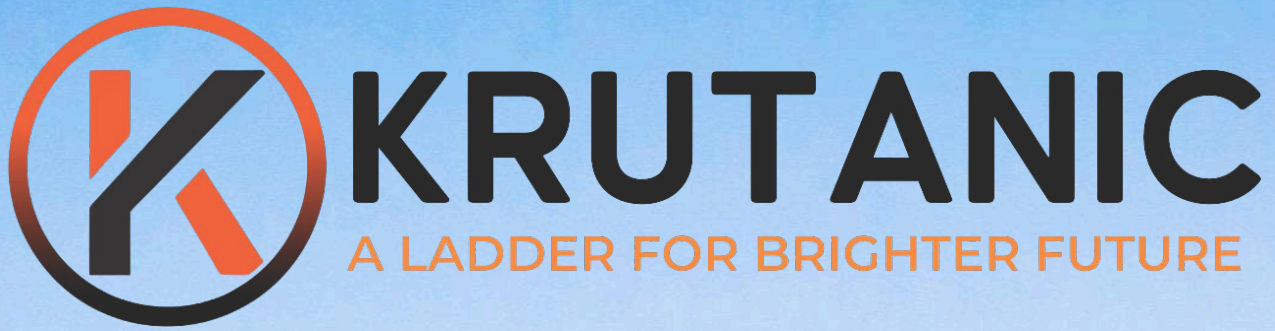
<https://www.krutanic.com/verify>
USER VERIFICATION CODE: 6880446cbf6aa824890024f3

ENROLLMENT VERIFICATION CODE: KRT_5678



Shrikant
Shrikant Rathod
Executive Director





Companies where our alumni thrive



Come join us now
and contact us now



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www.krutanic.com

Bangalore, Karnataka

Our process

Quick Guide

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Class

Certifications